THE BANACH–MAZUR DISTANCE BETWEEN PLANAR CONVEX BODIES IS AT MOST 11/5

MARIN VARIVODA

Abstract. We prove that the Banach–Mazur distance between any two planar convex bodies is at most 11/5, improving the bound $(19-\sqrt{73})/4 \approx 2.614$ of Brodiuk, Palko and Prymak. The conjectured maximum is $1+\sqrt{5}/2 \approx 2.118$, the distance between a triangle and a regular pentagon. The proof is computer assisted. We normalise a body by an inscribed affine-regular hexagon and sort all bodies into finitely many cells, each consisting of the bodies squeezed between two explicit rational polygons. The bound for every pair of cells is certified by rational witnesses, which a short independent program verifies in exact rational arithmetic. Along the way we obtain an explicit $\frac{1}{4}$-net of the planar Banach–Mazur compactum consisting of 6 729 polygons.

1. Introduction

For convex bodies $K, L \subset \mathbb{R}^2$ (compact convex sets with non-empty interior) the Banach–Mazur distance is

$$d(K,L) = \inf\bigl\{\lambda \geq 1 :\ T(L) \subseteq K \subseteq \lambda T(L) + b,\ T \text{ an invertible affine map},\ b \in \mathbb{R}^2\bigr\}.$$

It is symmetric and affine invariant, it satisfies $d(K,M) \leq d(K,L)\, d(L,M)$, and $\log d$ is a metric on the set of affine equivalence classes of planar convex bodies, which is compact, the *(planar) Banach–Mazur compactum*. We are interested in its diameter

$$D_2 = \sup\{d(K,L) :\ K, L \subset \mathbb{R}^2 \text{ convex bodies}\}.$$

By John's theorem [9] every planar convex body K satisfies $d(K, B^2) \leq 2$, hence $D_2 \leq 4$. Lassak [14] proved $D_2 \leq 3$, and Brodiuk, Palko and Prymak [4] proved $D_2 \leq \frac{19-\sqrt{73}}{4} \approx 2.614$; this is still listed as the best known planar bound in the recent work of Brandenberg, González Merino and Grundbacher [3] on the higher-dimensional problem. On the other hand, the distance between a triangle Δ and the regular pentagon P_5 is $1+\frac{\sqrt{5}}{2} \approx 2.118$ (Figure 1; see [7, 13, 10]), and Lassak conjectured that this is the diameter, $D_2 = 1+\frac{\sqrt{5}}{2}$ [13, 14, 15]. For centrally symmetric bodies the corresponding diameter is $\frac{3}{2}$, by a theorem of Stromquist [18], and for a symmetric body L and an arbitrary K one has $d(K,L) \leq 2$, with equality only for triangles [10]. In higher dimensions the order of the diameter of the compactum of not necessarily symmetric bodies is known up to logarithmic factors [17, 2].

Theorem 1.1. *For all planar convex bodies K and L,* $d(K,L) \leq \dfrac{11}{5}$.

In particular every planar convex body K contains a triangle T with $K \subseteq c + \frac{11}{5}(T-c)$ for some c; to our knowledge the best previous constant for this was $\frac{9}{4}$, due to Fleischer, Mehlhorn, Rote, Welzl and Yap [7].

Date: October 19, 2026.

2020 *Mathematics Subject Classification.* 52A10, 52A40, 46B20.

Key words and phrases. Banach–Mazur distance, planar convex bodies, Banach–Mazur compactum, ε-net, computer-assisted proof.

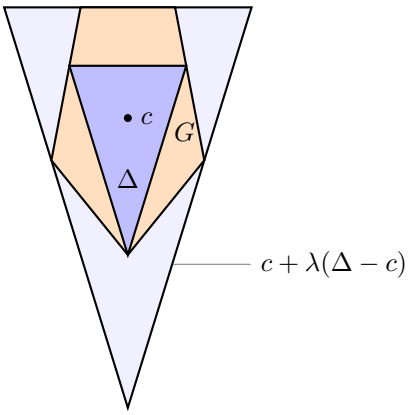


FIGURE 1. An optimal sandwich for a triangle and a regular pentagon: $\Delta \subseteq G \subseteq c + \lambda(\Delta - c)$ with G an affine-regular pentagon and $\lambda = 1 + \frac{\sqrt{5}}{2}$.

Method. The proof has a short geometric part and a large but elementary computational part. By a theorem of Besicovitch [1] every planar convex body K has an inscribed affine-regular hexagon; after an affine map it is a fixed regular hexagon H, and then K lies in the hexagram formed by H and the six equilateral triangles erected on its sides. In each of these triangles we use two slopes as coordinates; in them, the relevant geometry becomes a matter of comparing coordinates (Section 2). Six points over each side define a finite family of polygons $\mathcal{N}$ with 6 729 members up to symmetry, and every convex body is within distance $\frac{5}{4}$ of a member of $\mathcal{N}$ (Theorem 3.2). This already gives $D_2 \le \frac{25}{16} \max_{P,Q\in\mathcal{N}} d(P,Q)$.

For the bound $\frac{11}{5}$ we use the net in a sharper form. Recording which of the six points over each side lie in K places K in a *cell*, all of whose bodies K satisfy $P \subseteq K \subseteq O$ for two explicit rational polygons P and O (Section 4). A single linear map that fits the polygons of two cells into each other bounds $d(K, L)$ for all bodies K, L of these cells (Proposition 5.1). Bounds through 60 fixed "hub" polygons handle all but 1 339 of the 22 643 085 pairs of cells, and each remaining pair is handled by splitting its cells on new points until every piece has such a bound; in all, 433 821 witnesses are used. The witnesses are found in floating point and then rounded to rationals. The resulting certificate is checked by an independent program that recomputes all cells and verifies every witness in exact rational arithmetic (Section 5). The code and the certificate are available at `https://github.com/marin-varivoda/planar-bm-diameter`.

Related work. Zong [20] proposed a computer-assisted programme for Hadwiger's covering conjecture based on finite nets of the Banach–Mazur compactum, and proved that such nets exist for convex bodies in every dimension; see also [8]. The asymptotic size of optimal nets is governed by the metric entropy of the compactum, estimated by Bronshtein [5] (and in high dimensions by Pisier [16]); see also the discussion in Kobos and Swanepoel [11]. Chen, Lyu, Gao and Wu [6] constructed an explicit $\frac{1}{4}$-net of the *centrally symmetric* planar convex bodies from lattice polygons, and described an algorithm for general planar bodies. To our knowledge, the net of Theorem 3.2 is the first explicit ε-net of the compactum of all, not necessarily symmetric, planar convex bodies; it is built on the inscribed hexagon rather than on lattice polygons. Numerical computations have also guided work on other Banach–Mazur problems, for instance on the distance between the cube and the cross-polytope [19], whose values in dimensions 3 and 4 were then established in [12].

2. The hexagon frame

2.1. **Coordinates.** Since d is affine invariant we may use the coordinates $(x, y/\sqrt{3})$, in which all relevant points are rational. All figures are drawn to true shape. Let

$$H = \operatorname{conv}\{(\pm 1, 0), (\pm 2, -1), (\pm 1, -2)\}, \qquad I = (0, -1),$$

the regular hexagon with centre I, and let ρ be the rotation by 60° about I; in these coordinates $\rho(I + (x, y)) = I + \left(\frac{x-3y}{2}, \frac{x+y}{2}\right)$. The sides of H are $s_k = \rho^k(s_0)$, $k \in \mathbb{Z}/6$, where $s_0 = [B, C]$ with $B = (-1, 0)$ and $C = (1, 0)$. Write $B_k = \rho^k(B)$ and $C_k = \rho^k(C)$; then $B_k = C_{k+1}$. Over s_0 we erect the equilateral triangle $\Delta_0 = \operatorname{conv}\{B, C, A\}$, $A = (0, 1)$, and put $\Delta_k = \rho^k(\Delta_0)$.

Lemma 2.1 (Besicovitch [1])**.** *Every planar convex body K has an inscribed affine-regular hexagon, i.e. an affine image of H whose six vertices lie on ∂K.*

We call K *normalised* if all vertices of H lie on ∂K (so $H \subseteq K$). By Lemma 2.1 every convex body is affinely equivalent to a normalised one.

Lemma 2.2. *If K is normalised, then $K \subseteq H \cup \Delta_0 \cup \cdots \cup \Delta_5$.*

Proof. Let $p \in K \setminus H$. The sides of H whose lines separate p from H form a non-empty set of consecutive sides. If it contained two consecutive sides, say s_0 and s_1 meeting at B, then $p - B = -\alpha(C - B) - \beta(B_1 - B)$ with $\alpha, \beta > 0$, so B would be a positive convex combination of p, C and B_1, hence an interior point of K. So the line of only one side separates p from H, say that of s_0, and then $p \in \Delta_0$, because the lines of s_1 and s_5 meet at A. □

2.2. **Slopes.** For a point $q = (x, y)$ with $y > 0$ and $|x| < 1$ put

$$a(q) = \frac{y}{1 + x}, \qquad b(q) = \frac{y}{1 - x}.$$

In true shape, $\sqrt{3}\,a(q)$ and $\sqrt{3}\,b(q)$ are the tangents of the angles $\angle qBC$ and $\angle qCB$. The map $q \mapsto (a(q), b(q))$ is a bijection from $\Delta_0 \setminus s_0$ onto the *slope square* $(0, 1]^2$, with inverse

$$q(a, b) = \left(\frac{b - a}{a + b}, \frac{2ab}{a + b}\right),$$

and we put $q_k = \rho^k \circ q$, the point with slopes (a, b) over the side s_k. We order slope points componentwise: $z \le z'$ (z' *dominates* z) if $a \le a'$ and $b \le b'$. Since the triangle $\operatorname{conv}\{B, C, q\}$ is cut out by $y \ge 0$ and the lines Bq and Cq,

$$q(z) \in \operatorname{conv}\{B, C, q(z')\} \iff z \le z'. \tag{1}$$

For a normalised body K let $Z_k(K) = \{z \in (0, 1]^2 : q_k(z) \in K\}$ be the set of slopes of the points of K over s_k. By Lemma 2.2,

$$K = H \cup q_0(Z_0(K)) \cup \cdots \cup q_5(Z_5(K)), \tag{2}$$

and since $B_k, C_k \in K$, (1) and convexity show that each $Z_k(K)$ is a *down-set*: if $z \in Z_k(K)$ and $z' \le z$, then $z' \in Z_k(K)$. The last ingredient is a condition at the vertices of H, which lie on ∂K. Let

$$m(t) = \frac{1 - t}{1 + 3t} \qquad (0 \le t \le 1),$$

a decreasing bijection of $[0, 1]$ with $m(m(t)) = t$.

Lemma 2.3 (vertex condition)**.** *Let K be normalised, $z \in Z_k(K)$ and $z' \in Z_{k+1}(K)$. Then $a(z) \le m(b(z'))$, equivalently $a(z) + b(z') + 3a(z)b(z') \le 1$.*

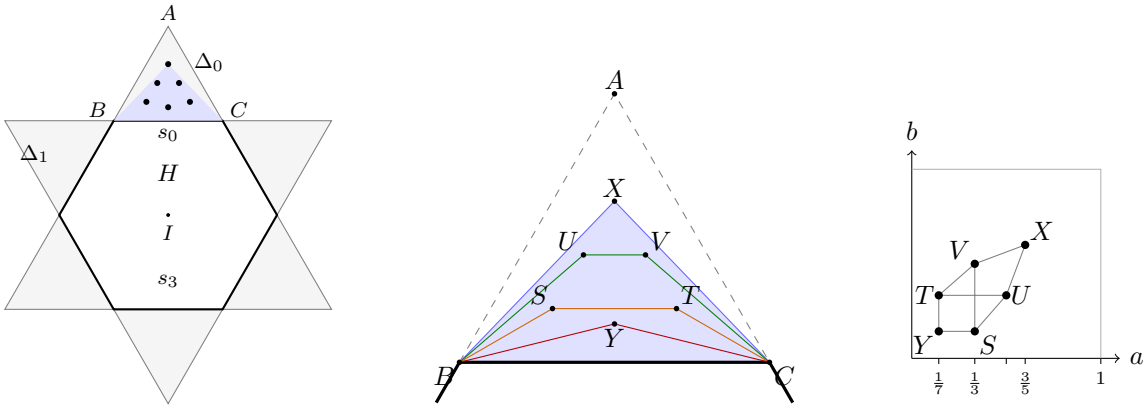


FIGURE 2. Left: the hexagon H and the hexagram $H \cup \Delta_0 \cup \cdots \cup \Delta_5$ containing every normalised body (Lemma 2.2); the net points over s_0. Middle: the triangle Δ_0 with the net points and some patches; shaded is the largest patch X. Right: the net points in slope coordinates, with the dominance order.

Proof. Work in true shape at the vertex $V = B_k = C_{k+1} \in \partial K$. The directions from V to $q_k(z)$, C_k, B_{k+1} and $q_{k+1}(z')$ occur in this angular order, with consecutive angles $\alpha = \arctan(\sqrt{3}a)$, $120°$ and $\beta = \arctan(\sqrt{3}b')$, where $a = a(z)$, $b' = b(z')$ and $\alpha, \beta \le 60°$. All four points lie in K, and a supporting line of K at V puts the four directions into a closed half-plane. Hence $\alpha + \beta \le 60°$, and $\sin(60° - \alpha - \beta) = \frac{\sqrt{3}}{2} \cos\alpha \cos\beta\,(1 - a - b' - 3ab')$. □

3. A $\frac{1}{4}$-NET

Over each side we use the six slope points

point	Y	S	T	U	V	X
slopes (a, b)	$(\frac{1}{7}, \frac{1}{7})$	$(\frac{1}{3}, \frac{1}{7})$	$(\frac{1}{7}, \frac{1}{3})$	$(\frac{1}{2}, \frac{1}{3})$	$(\frac{1}{3}, \frac{1}{2})$	$(\frac{3}{5}, \frac{3}{5})$
position over s_0	$(0, \frac{1}{7})$	$(-\frac{2}{5}, \frac{1}{5})$	$(\frac{2}{5}, \frac{1}{5})$	$(-\frac{1}{5}, \frac{2}{5})$	$(\frac{1}{5}, \frac{2}{5})$	$(0, \frac{3}{5})$

shown in Figure 2; for instance X is the image of A under the homothety $F(q) = I + \frac{4}{5}(q - I)$. In the dominance order $Y < S, T < U, V < X$, with S, T and U, V incomparable. For a normalised K the net points in $Z_k(K)$ form a down-set of the net. There are nine down-sets, the *patches*, which we name by their maximal elements:

$$0,\ Y,\ S,\ T,\ ST,\ U,\ V,\ UV,\ X.$$

For a patch p let $l(p)$ and $r(p)$ be the largest a- and b-slope of its points (0 for the empty patch 0). A *configuration* is a sequence $\mathbf{p} = (p_0, \ldots, p_5)$ of patches; it is *admissible* if $l(p_k) \le m(r(p_{k+1}))$ for all $k \in \mathbb{Z}/6$, and then $P(\mathbf{p})$ is the convex hull of H and of the points of p_k over s_k, $k = 0, \ldots, 5$. By Lemma 2.3, the configuration $\mathbf{p}(K)$ of the net points in a normalised body K is admissible.

Proposition 3.1. *There are* 77 625 *admissible configurations. The dihedral group* D_6 *of symmetries of* H *acts on them (a rotation shifts the indices, a reflection reverses them and exchanges* $S \leftrightarrow T$ *and* $U \leftrightarrow V$*), with* 6 729 *orbits.*

Proof. Enumeration (`proof.py`). As a check, by Burnside's lemma: the identity, the rotations by $\pm 60°$, $\pm 120°$, $180°$, the three reflections in lines through midpoints of opposite sides and the three in lines through opposite vertices fix 77 625, 5, 45, 275, 711 and 205 configurations each, and $\frac{1}{12}(77\,625 + 2 \cdot 5 + 2 \cdot 45 + 275 + 3 \cdot 711 + 3 \cdot 205) = 6\,729$. □

Theorem 3.2. *If K is normalised, then $P(\mathbf{p}(K)) \subseteq K \subseteq I + \frac{5}{4}\big(P(\mathbf{p}(K)) - I\big)$. Consequently the polygons $P(\mathbf{p})$, $\mathbf{p}$ admissible, form a $\frac{1}{4}$-net of the planar Banach–Mazur compactum: every planar convex body is within distance $\frac{5}{4}$ of one of them, and there are at most 6 729 of them up to affine equivalence.*

Proof. $P = P(\mathbf{p}(K)) \subseteq K$ by convexity. For the second inclusion we show $F(K) \subseteq P$ for $F(q) = I + \frac{4}{5}(q - I)$. As $F(H) \subseteq H$, by (2) and symmetry it suffices to show $F(q) \in P$ for $q \in K \cap \Delta_0 \setminus s_0$. All net points dominated by the slopes of q lie in K, hence in P, so it suffices that

$$F(q) \in \operatorname{conv}\big(H \cup \{\text{net points over } s_0 \text{ dominated by the slopes of } q\}\big) \qquad \text{for all } q \in \Delta_0 \setminus s_0.$$

Cut the slope square along $a, b \in \{\frac{1}{7}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}\}$ into 25 rectangles $[a_0, a_1] \times [b_0, b_1]$. For q in such a rectangle the right-hand side contains the polygon $P_{a_0 b_0}$ spanned by H and the net points $z \le (a_0, b_0)$. The points with slopes in the rectangle form the quadrilateral bounded by the lines through B with $a \in \{a_0, a_1\}$ and the lines through C with $b \in \{b_0, b_1\}$, with vertices $q(a, b)$, $a \in \{a_0, a_1\}$, $b \in \{b_0, b_1\}$ (a vertex with $a = 0$ or $b = 0$ is C or B, or both). As F is affine and $F(B), F(C) \in H$, it suffices to check $F(q(a, b)) \in P_{a_0 b_0}$ for the vertices with $a, b > 0$. These are finitely many exact point-in-polygon tests, which all succeed (`proof.py`). Several are tight, e.g. $F(A) = X$. □

Remark 3.3. By the multiplicative triangle inequality, Theorem 3.2 gives $D_2 \le \frac{25}{16} m^*$, where m^* is the largest distance between two members of the net. Numerically $m^* \approx 1.6145$, attained between $P(0, X, 0, X, 0, X)$, the member closest to a triangle, and $P(Y, Y, T, V, U, S)$; a certified bound of this kind would give $D_2 < 2.53$. We do not use this route: the proof of Theorem 1.1 uses the net points, but not the factor $\frac{5}{4}$.

4. Cells

A cell records some slope points that lie in a body and some that do not. It is then squeezed between two polygons.

Definition 4.1. A *cell* $\mathcal{C}$ consists of finite sets $I_k, E_k \subset (0, 1)^2$ of slope points, $k \in \mathbb{Z}/6$ ("inside" and "outside"). A normalised body K *belongs* to $\mathcal{C}$ if $I_k \subseteq Z_k(K)$ and $E_k \cap Z_k(K) = \emptyset$ for all k. The *inner polygon* $P(\mathcal{C})$ is the convex hull of H and of the points $q_k(z)$, $z \in I_k$. Let l_k and r_k be the largest a- and b-slope in I_k (0 if $I_k = \emptyset$), and let the *staircase* S_k be the set of $(a, b) \in [0, m(r_{k+1})] \times [0, m(l_{k-1})]$ such that $a \le a_e$ or $b \le b_e$ for every $(a_e, b_e) \in E_k$. The *outer polygon* $O(\mathcal{C})$ is the convex hull of H and of the points $q_k(c)$ for the corners c of the staircases S_k (Figure 3).

The *corners* of S_k are its maximal points; every point of S_k is dominated by a corner. Their coordinates are among $m(r_{k+1})$ and the a_e, resp. $m(l_{k-1})$ and the b_e, so there are finitely many.

Lemma 4.2. *If K belongs to the cell $\mathcal{C}$, then $P(\mathcal{C}) \subseteq K \subseteq O(\mathcal{C})$.*

Proof. The first inclusion holds by convexity. For the second, by (2) it suffices to show that every $w = (a, b) \in Z_k(K)$ lies in S_k: then w is dominated by a corner c, and $q_k(w) \in \operatorname{conv}\{B_k, C_k, q_k(c)\} \subseteq O(\mathcal{C})$ by (1). As $Z_k(K)$ is a down-set that does not contain any $e \in E_k$, w does not dominate e, so $a < a_e$ or $b < b_e$. And by Lemma 2.3, applied to w and a point of I_{k+1} with b-slope r_{k+1}, and to a point of I_{k-1} with a-slope l_{k-1} and w, we have $a \le m(r_{k+1})$ and $b \le m(l_{k-1})$. □

Every admissible configuration $\mathbf{p}$ defines a *root cell* $\mathcal{C}(\mathbf{p})$: I_k consists of the net points of p_k and E_k of the other net points. Its inner polygon is $P(\mathbf{p})$. Figure 4 shows two root cells.

Lemma 4.3. *Every planar convex body is affinely equivalent to a body that belongs to one of the 6 729 root cells of the representatives of Proposition 3.1.*

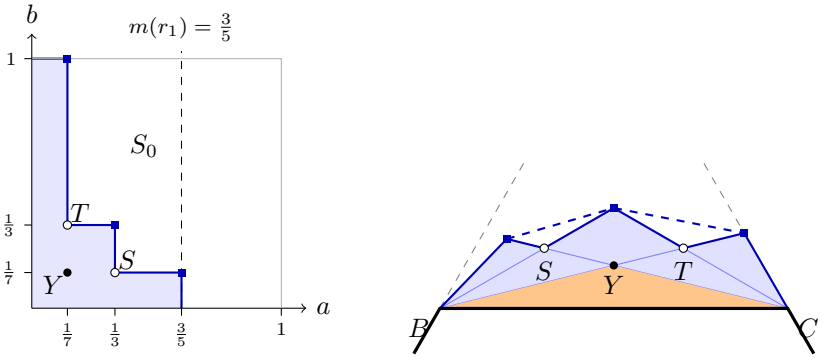


FIGURE 3. The staircase S_0 for $I_0 = \{Y\}$, $E_0 = \{S, T\}$, $r_1 = \frac{1}{7}$ and $l_5 = 0$ (left), and its image over the side s_0 (right): the part over s_0 of every body of the cell lies in the union (light) of the triangles $\operatorname{conv}\{B, C, q(c)\}$ over the three corners c (squares), and it contains the triangle $\operatorname{conv}\{B, C, q(Y)\}$ (dark). The outer polygon uses the convex hull (dashed).

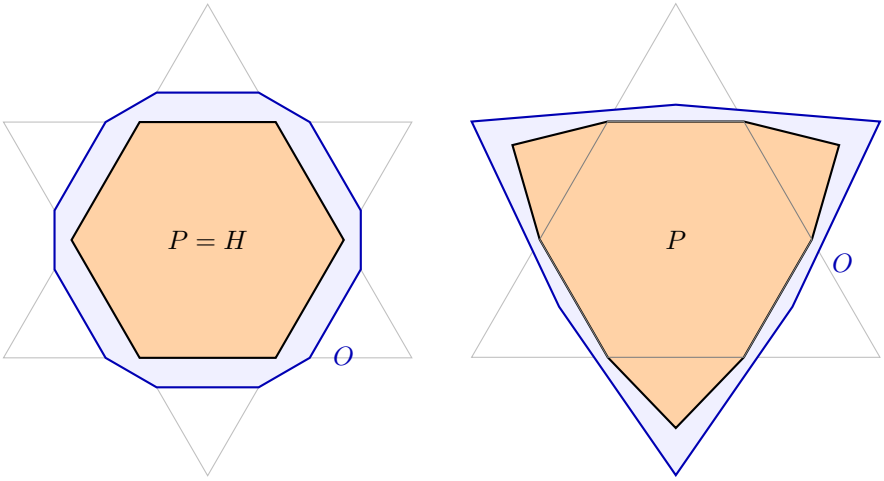


FIGURE 4. The root cells of the configurations $(0, 0, 0, 0, 0, 0)$ (left) and $(0, X, 0, X, 0, X)$ (right): every normalised body K whose net points are exactly those of the configuration satisfies $P \subseteq K \subseteq O$. The right cell contains the triangle $\operatorname{conv}\{(-3, 0), (3, 0), (0, -3)\}$.

Proof. We may assume that K is normalised; it belongs to $\mathcal{C}(\mathbf{p}(K))$, and $\mathbf{p}(K)$ is admissible. A symmetry $g \in D_6$ of H fixes I, maps Δ_k to $\Delta_{g(k)}$ and preserves the slopes, except that reflections exchange a and b. So $g(K)$ is normalised and belongs to $\mathcal{C}(g\mathbf{p}(K))$, with D_6 acting on configurations as in Proposition 3.1. Choose g with $g\mathbf{p}(K)$ the representative of its orbit. □

A cell can be refined. *Splitting* the cell $\mathcal{C}$ on a slope point z over the side s_k gives the two cells $\mathcal{C}^+$ and $\mathcal{C}^-$ obtained by adding z to I_k, resp. to E_k. Every body of $\mathcal{C}$ belongs to $\mathcal{C}^+$ or to $\mathcal{C}^-$.

5. The proof

Proposition 5.1. *Let $\mathcal{C}$, $\mathcal{D}$ be cells, M a linear map, $t, s \in \mathbb{R}^2$ and $\lambda > 0$ with*

$$M\,O(\mathcal{D}) + t \subseteq P(\mathcal{C}) \qquad \text{and} \qquad O(\mathcal{C}) \subseteq \lambda M\,P(\mathcal{D}) + s.$$

Then $d(K, L) \le \lambda$ for every body K of $\mathcal{C}$ and every body L of $\mathcal{D}$.

root cells, hub polygons	6 729, 60
hub witnesses	403 740
pairs of root cells (diagonal included)	22 643 085
pairs not covered by hubs (proof trees)	1 339
splits, witnesses in the proof trees	28 742, 30 081
largest witness value in the proof trees	2.1999978 . . .
certificate size (compressed)	22 MB

TABLE 1. The certificate for Theorem 1.1.

Proof. M is invertible, as $\lambda M P(\mathcal{D})$ contains a translate of $O(\mathcal{C})$, which has interior points. By Lemma 4.2, $ML + t \subseteq M\,O(\mathcal{D}) + t \subseteq P(\mathcal{C}) \subseteq K \subseteq O(\mathcal{C}) \subseteq \lambda M\,P(\mathcal{D}) + s \subseteq \lambda ML + s$, and $\lambda ML + s = \lambda(ML + t) + (s - \lambda t)$. □

We call (M, t, s, λ) a *witness* of value λ for the pair $(\mathcal{C}, \mathcal{D})$. The same holds with a fixed convex polygon Q in place of $\mathcal{D}$, with $P(\mathcal{D}) = O(\mathcal{D}) = Q$ and $L = Q$. All polygons involved have rational vertices, so a rational witness can be verified exactly: it suffices to check that each vertex of $M\,O(\mathcal{D}) + t$ lies in $P(\mathcal{C})$ and that each vertex of $O(\mathcal{C})$ lies in $\lambda M P(\mathcal{D}) + s$.

Witnesses for all of the 22 643 085 pairs of root cells would be costly, and some pairs need more than one witness. We use two devices. The first is the triangle inequality: if $d(K, Q) \le e$ and $d(L, Q) \le e'$, then $d(K, L) \le ee'$. The second is splitting: we call a *proof tree* for a pair of cells $(\mathcal{C}, \mathcal{D})$ either a witness of value at most $\frac{11}{5}$ for $(\mathcal{C}, \mathcal{D})$, or a split of $\mathcal{C}$ (or of $\mathcal{D}$) on a slope point in $(0, 1)^2$ into two cells $\mathcal{C}^{\pm}$, together with proof trees for both pairs $(\mathcal{C}^+, \mathcal{D})$ and $(\mathcal{C}^-, \mathcal{D})$ (resp. $(\mathcal{C}, \mathcal{D}^{\pm})$). Since every body of a cell belongs to one of its two halves, Proposition 5.1 and induction give:

Lemma 5.2. *If a pair of cells $(\mathcal{C}, \mathcal{D})$ has a proof tree, then $d(K, L) \le \frac{11}{5}$ for all bodies K of $\mathcal{C}$ and L of $\mathcal{D}$.*

5.1. **The certificate.** Let $\mathcal{C}_1, \dots, \mathcal{C}_n$, $n = 6\,729$, be the root cells. The certificate fixes 60 *hub* polygons $Q_1, \dots, Q_{60}$ (inner polygons of root cells: the hexagon, the member closest to the triangle, two other symmetric members, and a sample of others spread out with respect to a simple affine invariant; any choice of hubs would be valid) and contains

(1) for every root cell $\mathcal{C}_i$ and every hub Q_h, a rational witness, of value $e_{i,h}$;
(2) for every pair $i \le j$ with $e_{i,h}e_{j,h} > \frac{11}{5}$ for all hubs h, a proof tree for $(\mathcal{C}_i, \mathcal{C}_j)$, with rational witnesses and split points.

The checker (`proof.py`) recomputes the root cells, verifies every witness in exact rational arithmetic, follows the proof trees by recomputing the split cells, and determines the pairs that need a proof tree. Table 1 summarises the certificate.

Proof of Theorem 1.1. Let K and L be planar convex bodies. By Lemma 4.3 we may assume that K belongs to the root cell $\mathcal{C}_i$ and L to the root cell $\mathcal{C}_j$, and since d is symmetric, that $i \le j$. If $e_{i,h}e_{j,h} \le \frac{11}{5}$ for some hub h, then $d(K, L) \le d(K, Q_h)\,d(Q_h, L) \le \frac{11}{5}$ by Proposition 5.1. Otherwise the certificate contains a proof tree for $(\mathcal{C}_i, \mathcal{C}_j)$, and $d(K, L) \le \frac{11}{5}$ by Lemma 5.2. □

5.2. **How the certificate was found.** None of this needs to be correct, as only the final verification counts. For a linear map M let μ be the largest number with $\mu M\,O(\mathcal{D}) + t \subseteq P(\mathcal{C})$ for some t, and σ the largest with $\sigma\,O(\mathcal{C}) + s \subseteq M\,P(\mathcal{D})$ for some s; both are linear programs in three variables. Then $(\mu M, t, -s/\sigma, 1/(\mu\sigma))$ is a witness, and its value does not change when M is scaled. We minimise it over $M = W_{\mathcal{C}}^{-1} R_\theta \left(\begin{smallmatrix} e^{\alpha} & \beta \\ 0 & e^{-\alpha} \end{smallmatrix}\right) \operatorname{diag}(1, \pm 1) W_{\mathcal{D}}$, where $W_{\mathcal{C}}, W_{\mathcal{D}}$ bring $P(\mathcal{C}), P(\mathcal{D})$ to isotropic position, by the Nelder–Mead method started from the best 32 of 4 096 random parameters. The witness is then shrunk (inside) and

enlarged (outside) by a relative 10^{-8} about the image of I, and rounded to rationals with denominator 10^{10}. Proof trees are grown greedily: if the witness of a pair has value above $\frac{11}{5}$, we split one of the two cells at the outer corner c that is active in the witness (it lies on the boundary of $\lambda MP(\mathcal{D}) + s$, resp. is mapped to the boundary of $P(\mathcal{C})$) and farthest from the inner polygon; the split point is $\frac{1+\tau}{2}c$ (rounded), where τc is the largest multiple of c dominated by a point of I_k. The search took about 45 minutes on 30 cores for the hub witnesses and 11 minutes on 20 cores for the proof trees, and the checker accepts the certificate in under a minute on 20 cores.

6. Remarks

Remark 6.1 (the root level). Without splitting, direct witnesses for all root pairs give a bound of about 2.39. The worst pairs involve the root cell $\mathcal{C}(0, X, 0, X, 0, X)$ of Figure 4, which contains a triangle together with polygons of the net that are at distance $\frac{5}{4}$ from it, partnered with cells of nearly centrally symmetric bodies. Already the hexagon cell and this cell show the effect of the outer polygons: the rational witness

$$M = \tfrac{1}{6}\left(\begin{smallmatrix} 3 & 3 \\ -1 & 3 \end{smallmatrix}\right), \quad t = \left(\tfrac{1}{2}, -\tfrac{1}{2}\right), \quad s = \left(\tfrac{9}{8}, \tfrac{1}{8}\right), \quad \lambda = \tfrac{9}{4}$$

for $\mathcal{C}(0, 0, 0, 0, 0, 0)$ against $\mathcal{C}(0, X, 0, X, 0, X)$ gives $\frac{9}{4}$, the square of $d(H, \Delta) = \frac{3}{2}$: here the looseness of the two cells costs a whole factor $\frac{3}{2}$.

Remark 6.2 (lower targets). The same refinement at the targets 2.15 and 2.18 did not terminate in reasonable time: each further level of splitting lowered the largest pair bound by only about 0.005, while multiplying the number of pending pairs by about 3. Since $d(\Delta, S) = 2$ for every centrally symmetric body S (the asymmetry of Δ is 2, and see [10]), a target close to $1 + \frac{\sqrt{5}}{2}$ leaves very little room for the looseness of both cells of a pair involving the triangle. For centrally symmetric bodies one could use instead a concentric inscribed affine-regular hexagon H with $K \subseteq \frac{3}{2}H$ (Lassak [13]), which truncates every cap; for general bodies near the triangle, a second chart, for instance a normalisation by a maximal inscribed triangle, might help.

References

[1] A. S. Besicovitch, *Measure of asymmetry of convex curves*, J. London Math. Soc. **23** (1948), 237–240.

[2] P. Bizeul and B. Klartag, *Distances between non-symmetric convex bodies: optimal bounds up to polylog*, preprint, arXiv:2510.20511 (2025).

[3] R. Brandenberg, B. González Merino, and F. Grundbacher, *Tight stability estimates near the simplex and applications for the Banach–Mazur distance and Rogers–Shephard-type inequalities*, preprint, arXiv:2607.27041 (2026).

[4] S. Brodiuk, N. Palko, and A. Prymak, *On Banach–Mazur distance between planar convex bodies*, Aequationes Math. **92** (2018), 993–1000.

[5] E. M. Bronshtein, *ε-entropy of convex sets and functions*, Siberian Math. J. **17** (1976), 393–398.

[6] Y. Chen, Y. Lyu, S. Gao, and S. Wu, *Construction of ε-nets for the space of planar convex bodies endowed with the Banach–Mazur metric*, Mathematics **13** (2025), 1358.

[7] R. Fleischer, K. Mehlhorn, G. Rote, E. Welzl, and C. Yap, *Simultaneous inner and outer approximation of shapes*, Algorithmica **8** (1992), 365–389.

[8] S. Gao, H. Martini, S. Wu, and L. Zhang, *New covering and illumination results for a class of polytopes*, Arch. Math. **122** (2024), 599–607.

[9] F. John, *Extremum problems with inequalities as subsidiary conditions*, in: Studies and Essays Presented to R. Courant on his 60th Birthday, Interscience, New York, 1948, 187–204.

[10] T. Kobos, *Extremal Banach–Mazur distance between a symmetric convex body and an arbitrary convex body on the plane*, Mathematika **66** (2020), 161–177.

[11] T. Kobos and K. J. Swanepoel, *Equilateral dimension of the planar Banach–Mazur compactum*, Proc. Amer. Math. Soc. **153** (2025), 4423–4436.

[12] T. Kobos and M. Varivoda, *On the Banach–Mazur distance in small dimensions*, Discrete Comput. Geom. **74** (2025), 399–427.

[13] M. Lassak, *Approximation of convex bodies by triangles*, Proc. Amer. Math. Soc. **115** (1992), 207–210.

[14] M. Lassak, *Banach–Mazur distance of planar convex bodies*, Aequationes Math. **74** (2007), 282–286.

[15] M. Lassak, *Banach–Mazur distance between convex quadrangles*, Demonstratio Math. **47** (2014), 989–993.
[16] G. Pisier, *On the metric entropy of the Banach–Mazur compactum*, Mathematika **61** (2015), 179–198.
[17] M. Rudelson, *Distances between non-symmetric convex bodies and the MM^*-estimate*, Positivity **4** (2000), 161–178.
[18] W. Stromquist, *The maximum distance between two-dimensional Banach spaces*, Math. Scand. **48** (1981), 205–225.
[19] F. Xue, *On the Banach–Mazur distance between the cube and the crosspolytope*, Math. Inequal. Appl. **21** (2018), 931–943.
[20] C. Zong, *A quantitative program for Hadwiger's covering conjecture*, Sci. China Math. **53** (2010), 2551–2560.

Department of Mathematics, Faculty of Science, University of Zagreb, Bijenička cesta 30, 10000 Zagreb, Croatia
Email address: marin.varivoda@math.hr